Data Interpretation Exercise

Data Interpretation Assessment (R Programming)

Topics:

- Mapping Data onto Aesthetics
- Coordinate Systems and Axes
- Color Scales

Instructions: Use the given dataset to write an appropriate R program, generate the required visualization, and answer the questions.

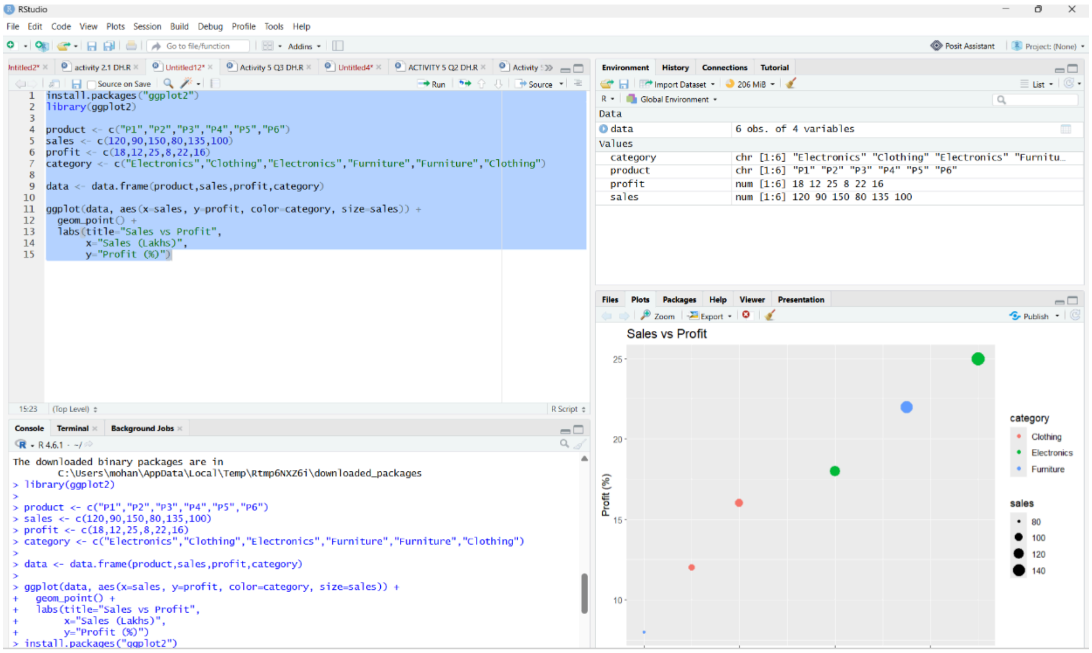
Question 1: Mapping Data onto Aesthetics

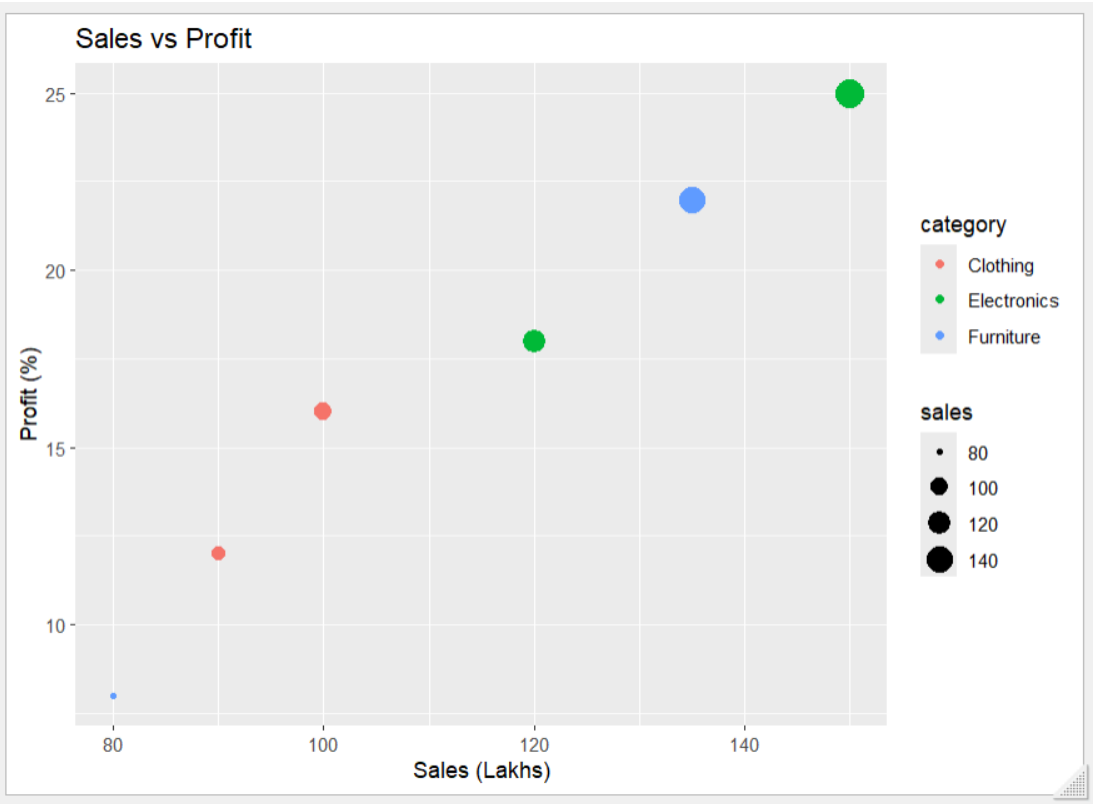
Dataset

Product	Sales (₹ Lakhs)	Profit (%)	Category
P1	120	18	Electronics
P2	90	12	Clothing
P3	150	25	Electronics
P4	80	8	Furniture
P5	135	22	Furniture
P6	100	16	Clothing

Questions

1. Write an R program to create a scatter plot with **Sales** on the x-axis and **Profit** on the y-axis.





2. Map **Category** using an appropriate aesthetic and **Sales** using another suitable aesthetic.

Answer: Category → Color

Sales → Size

3. Interpret the relationship between Sales and Profit.
   Answer:
   There is a positive relationship between Sales and Profit. Higher sales generally result in higher profit.
4. Identify the type of scale used for each variable.
   Types of scale:

- Sales → Continuous Scale
- Profit → Continuous Scale
- Category → Categorical (Discrete) Scale

5. Suggest one alternative visualization that better represents the same data.
   Answer: Bubble Chart.

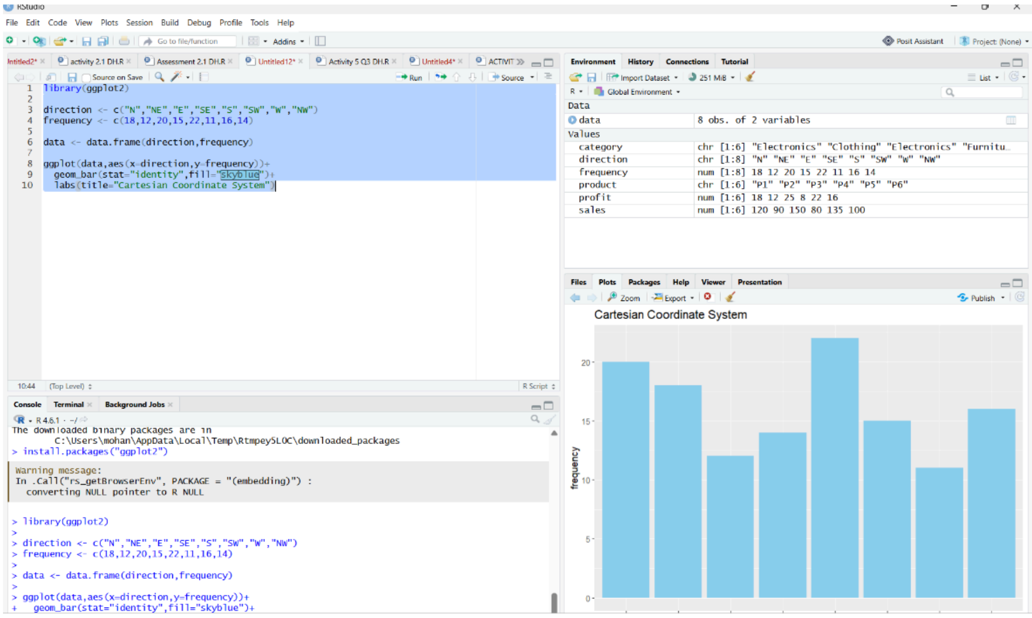
Question 2: Coordinate Systems (Cartesian and Polar)

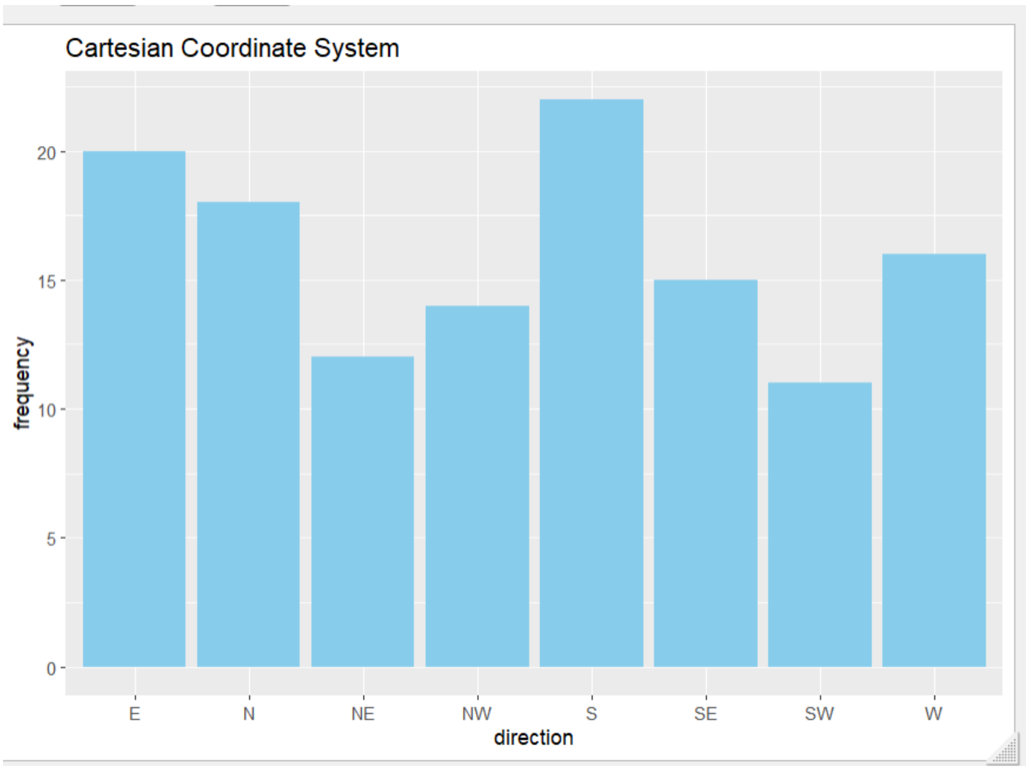
Dataset

Direction	Frequency
N	18
NE	12
E	20
SE	15
S	22
SW	11
W	16
NW	14

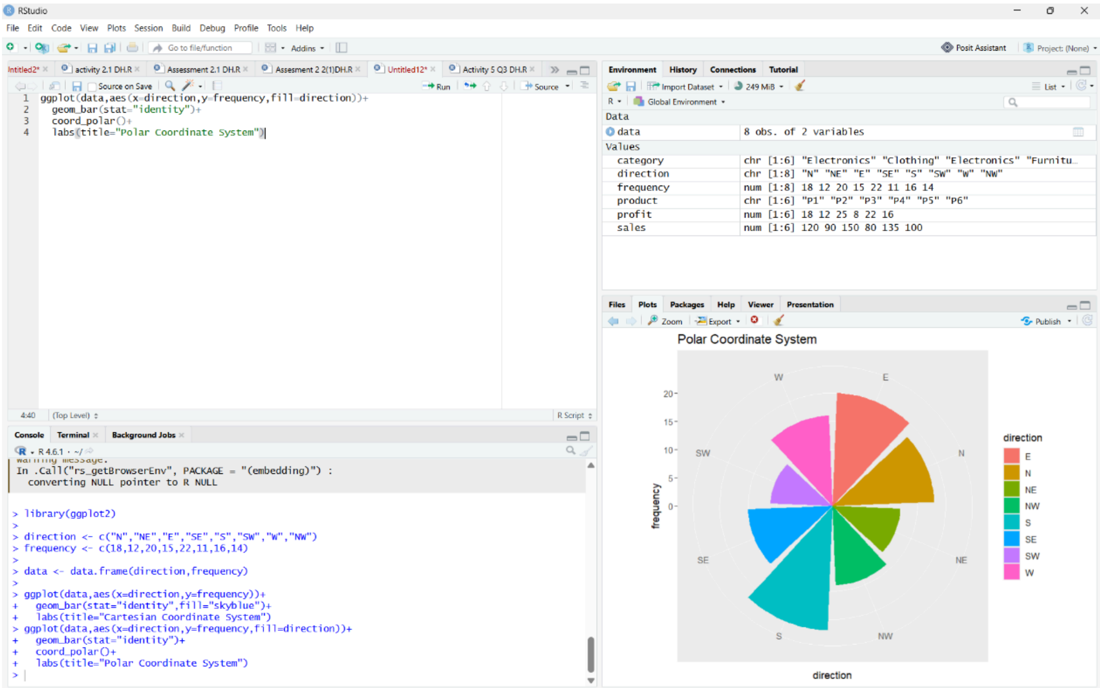
Questions

1. Write an R program to visualize the dataset using a Cartesian coordinate system.





2. Write another R program to visualize the same data using a Polar coordinate system.



2. Compare the effectiveness of both visualizations.

Answer:

- Cartesian is easier for comparing values.
- Polar is better for showing directional patterns.

4. Explain which coordinate system is more appropriate for directional data.

Answer: Polar Coordinate System.

5.Justify your answer based on data interpretation.

Answer: Directional data naturally follows a circular arrangement, making polar coordinates easier to interpret.

Question 3: Color Scales for Data Representation

Dataset

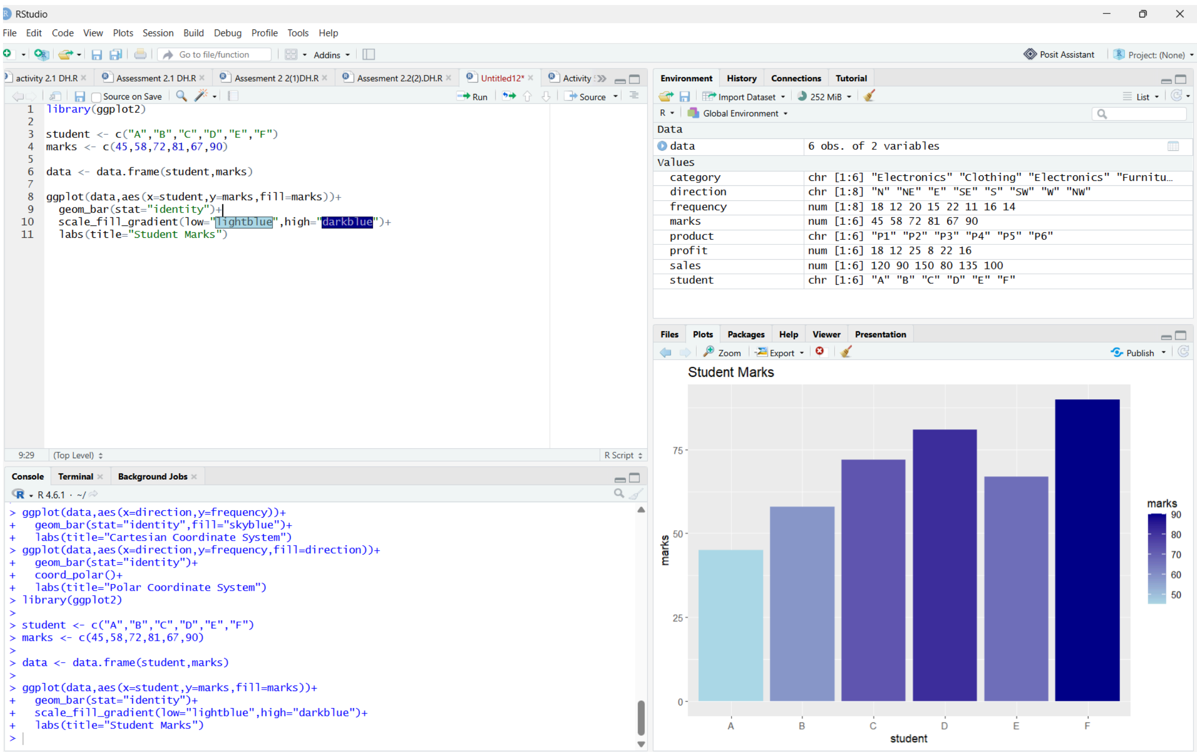
Student	Marks

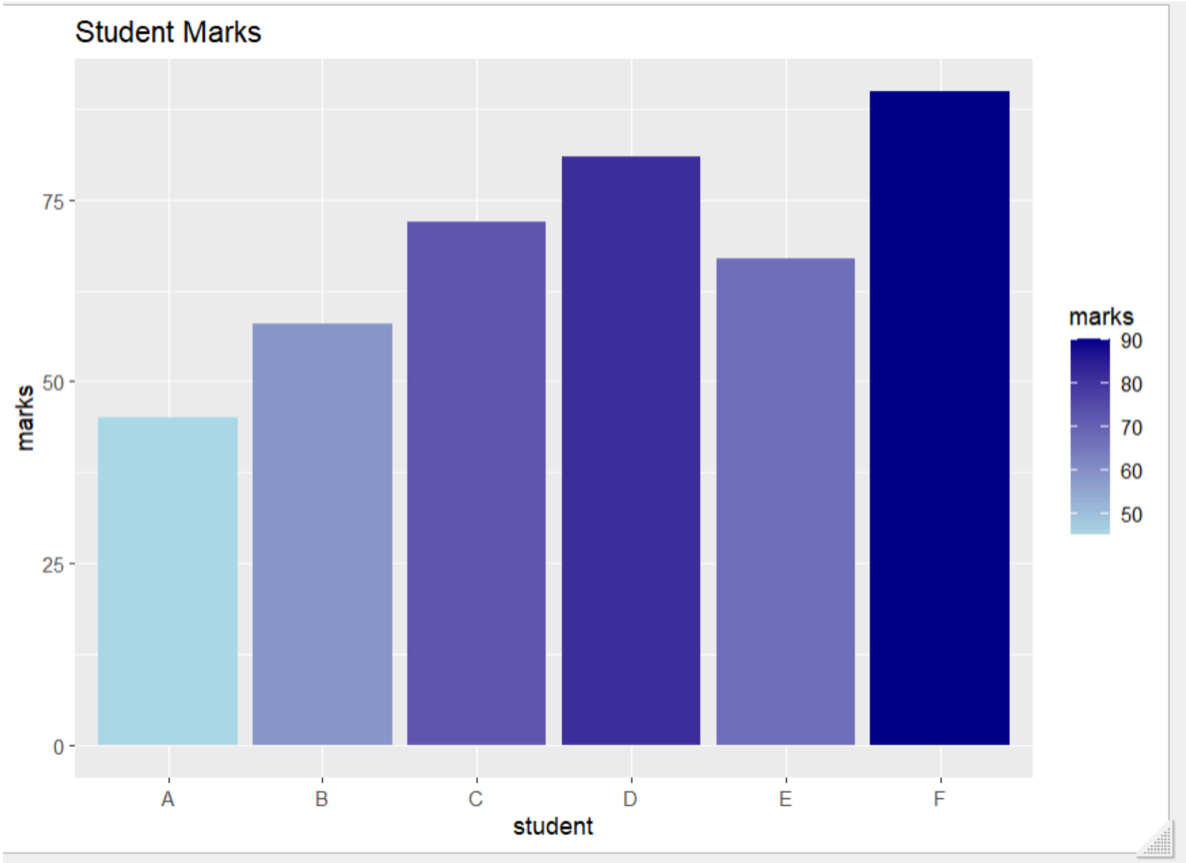
A	45
B	58
C	72

Student	Marks
D	81
E	67
F	90

Questions

1. Write an R program to create a bar chart using a sequential color scale based on Marks.





2. Identify the students with the highest and lowest marks from the visualization.

Highest and Lowest Marks:

- **Highest → Student F (90)**
- **Lowest → Student A (45)**

3.Explain why a sequential color scale is appropriate.

Answer: It represents increasing numerical values using gradually changing colors.

4.Suggest a color palette suitable for color-blind users.
Answer: Viridis.

5.Explain the difference between sequential and categorical color scales.

Answer: Difference between Sequential and Categorical scales
Sequential Scale

- Used for continuous numerical values.
- Color intensity changes gradually.

Categorical Scale

- Used for categories.
- Different colors represent different groups.

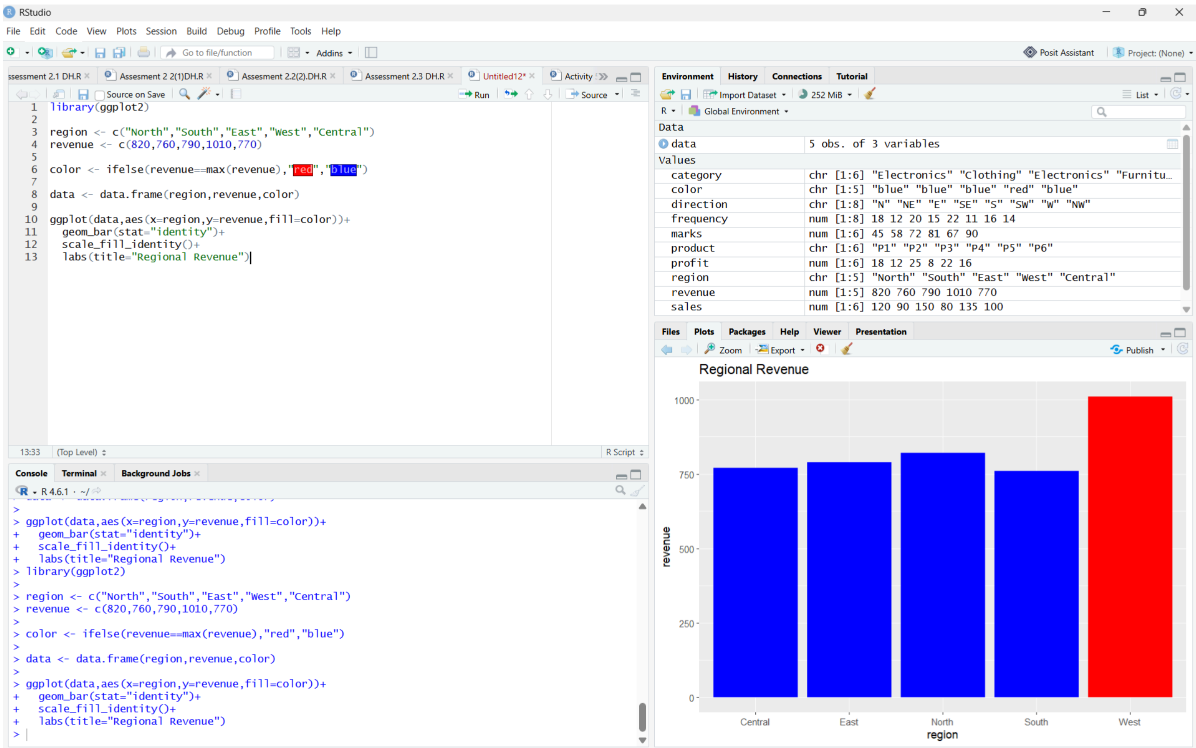
Question 4: Highlighting with Color

Dataset

Region	Revenue (₹ Crores)
North	820
South	760
East	790
West	1010
Central	770

Questions

1. Write an R program to create a bar chart highlighting the region with the highest revenue.



2. Explain how highlighting improves data interpretation.
   Answer: Highlighting makes the most important data stand out immediately.
3. Identify the highest-performing region.
   Answer: West (₹1010 Crores).
4. Discuss how excessive use of colors can affect interpretation.
   Answer : Too many colors can confuse viewers and reduce readability.
5. Suggest one alternative highlighting technique other than color.
   Answer: Use labels, arrows, bold borders, or annotations.

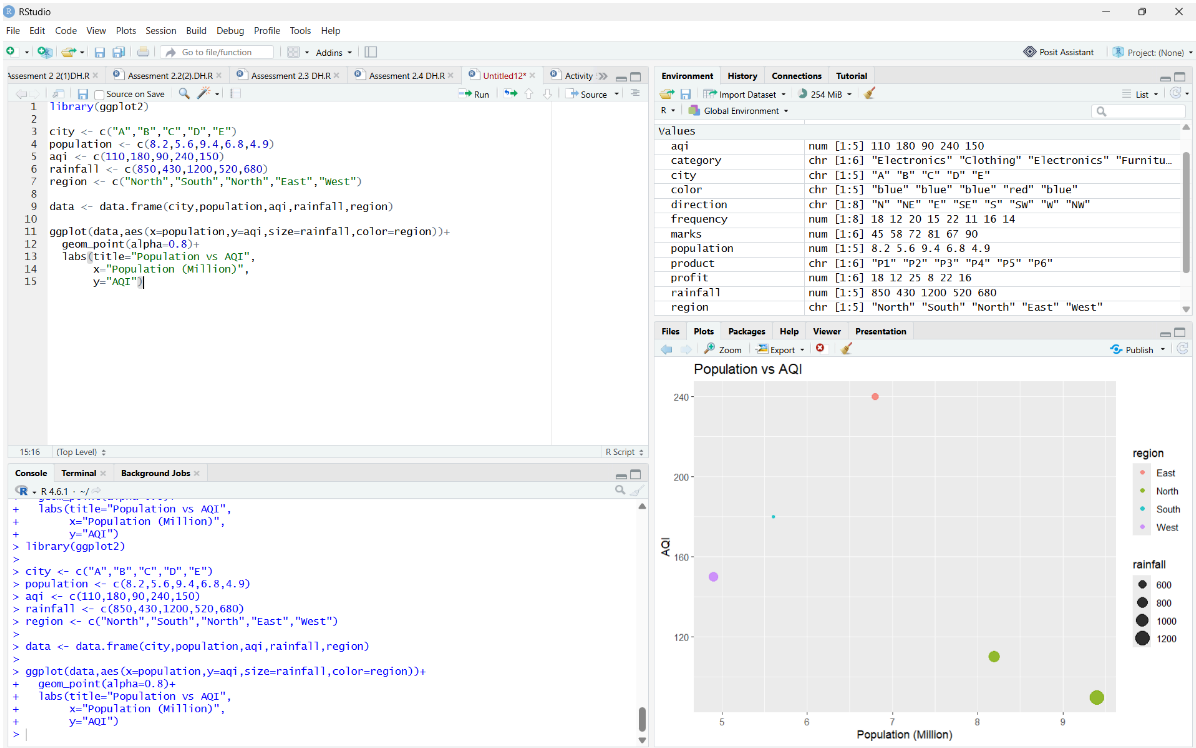
Question 5: Integrated Visualization

Dataset

City Population (Rainfall			North South North East West
8.2	110	850	
5.6	180	430	
9.4	90	1200	
6.8	240	520	
4.9	150	680	

Questions

1. Write an R program to create a bubble chart using appropriate aesthetic mappings.



2. Select suitable aesthetics for Population, AQI, Rainfall, and Region.

Answer:

- Population → X-axis
- AQI → Y-axis
- Rainfall → Bubble Size
- Region → Color

3. Interpret the relationship between Population and AQI.

Answer: Cities with larger populations generally have higher AQI values, indicating more air pollution.

4. Identify the continuous and categorical variables.

Answer:

Continuous Variables

- Population
- AQI
- Rainfall

Categorical Variable

- Region

5. Recommend one improvement to enhance the visualization's readability.

Answer: Use data labels and a color-blind-friendly palette (Viridis) to improve readability.